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Bulletin of Indonesian Economic Studies

Publication details, including instructions for authors and subscription information:

<http://www.tandfonline.com/loi/cbie20>

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J. Vernon Henderson ^a & Ari Kuncoro ^b

^a Brown University, Providence RI
National Bureau of Economic Research ,
USA

^b University of Indonesia

Published online: 19 Aug 2006.

To cite this article: J. Vernon Henderson & Ari Kuncoro (1996) The Dynamics of Jabotabek Development, Bulletin of Indonesian Economic Studies, 32:1, 71-95, DOI: [10.1080/00074919612331336898](https://doi.org/10.1080/00074919612331336898)

To link to this article: <http://dx.doi.org/10.1080/00074919612331336898>

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THE DYNAMICS OF JABOTABEK DEVELOPMENT

J. Vernon Henderson*

*Brown University, Providence RI
National Bureau of Economic Research, USA*

Ari Kuncoro

University of Indonesia

Damhuri Nasution

University of Indonesia

This paper examines the dynamics of urban development of the Jakarta metropolitan region, called Jabotabek, from 1980 through the early 1990s. We focus on the rapidly changing spatial allocation of residences and business, finding that Jabotabek is following expected developmental patterns but faces special and some critical problems. Population densities remain unusually high and urban development is hindered by very poor land market institutions—weakly defined property rights particularly for traditional low income residents, complete lack of active land use planning, and relatively low infrastructure investments. Industry is rapidly suburbanising to take advantage of low land prices and wages in suburban Botabek. With toll road construction east and west from the city, within the five-year period 1986–1991, Jabotabek moved from being a monocentric city where core city industrial activity dominated to a predominantly multi-centred city. These developments and implicit government policy have hurt small-scale industry, which ultimately will hinder the whole development process. Issues of spatial mismatch, where low income workers' residences and workplaces become inordinately spatially separated due to government policies in land markets, herald increasing problems of urban underemployment and social unrest.

*The authors acknowledge the helpful comments of two anonymous referees.

The urbanised population of the metropolitan area of Jakarta, known as Jabotabek, grew at an average rate of 5.9% per year in 1980–1990.¹ For such a large metropolis, that is a very rapid growth rate, both in international terms and relative to other metro areas in Indonesia (World Bank 1993). The resulting centralisation of urban resources is discussed in Henderson and Kuncoro (1996), where the implicit subsidisation of industrial location in Jabotabek and the two or three other largest metro areas in Indonesia is analysed.

The present paper focuses on Jabotabek's spatial development in response to this massive infusion of population and the corresponding investment in infrastructure. Jabotabek is responding to urban growth with the same patterns of population and industrial decentralisation and suburbanisation that we are familiar with from other countries (see, for example, Ingram and Carroll 1981). We analyse these patterns and point out some unusual features: rather high population densities near the centre combined with urban sprawl at the fringes; rapid suburbanisation of formal sector manufacturing; and rapidly changing patterns of land use. We explore these features by looking at the evolving spatial distribution of both population and industrial activity, discussing reasons for Jabotabek's unusual features and pointing out the policy-institutional questions they raise.

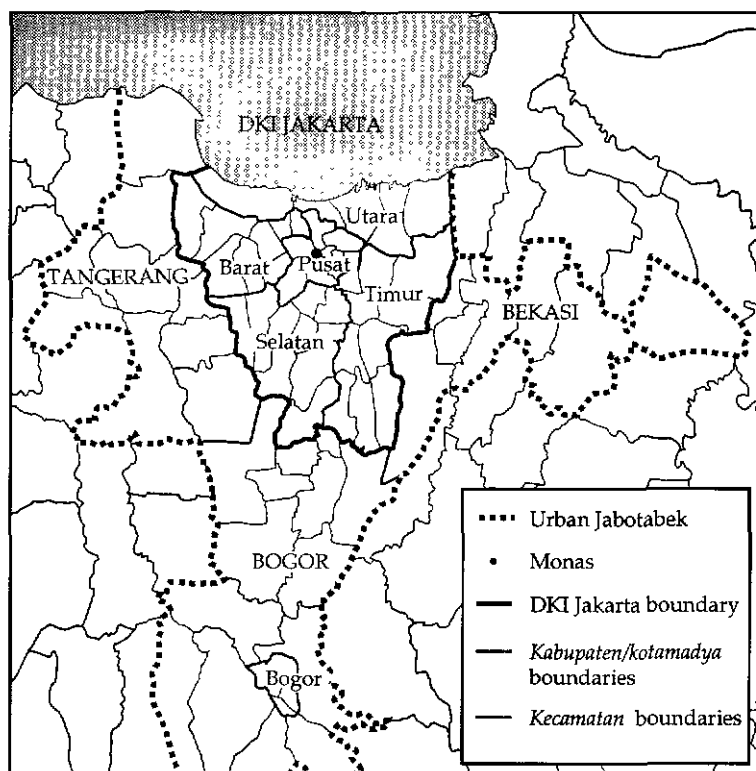
POPULATION DISPERSION

Density and Land Use Change

In map 1, we picture the Jabotabek area, so the reader has in mind the basic layout of the metropolitan region for reference throughout the paper. Jakarta is a coastal city, with a major harbour. The heavy dotted line in the map is the 1990 boundary of urban Jabotabek;² the very heavy solid line indicates the borders of the core city of DKI Jakarta,

¹The acronym 'Jabotabek' is formed from the names of Jakarta and the neighbouring *kabupaten* of Bogor, Tangerang and Bekasi. The term 'Botabek' is used for the areas of Jabotabek that lie outside the Special Region (DKI) of Jakarta.

²Our sample includes all *kecamatan* with significant population densities and is based on the data and analysis of population densities at the *desa* level for 1990 in GOI (1993, figure 1.2). However, the *kecamatan* boundaries in map 1 are 1986 boundaries, so as to correspond to those in map 2, which necessarily uses 1986 boundaries. Between 1986 and 1990 a number of *kecamatan* were split up.

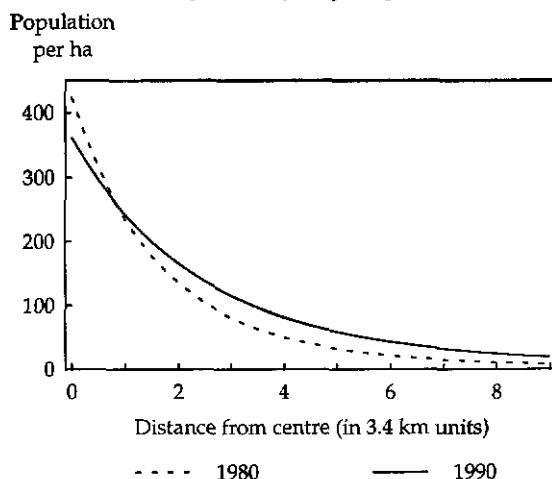
MAP 1 *Jabotabek*^a

^aKecamatan boundaries shown are those in use at the time of the 1986 Economic Census.

while those of the Central Business Area of Jakarta Pusat, around the National Monument (Monas), and of other *kabupaten* and *kotamadya*, are marked by the medium solid line.

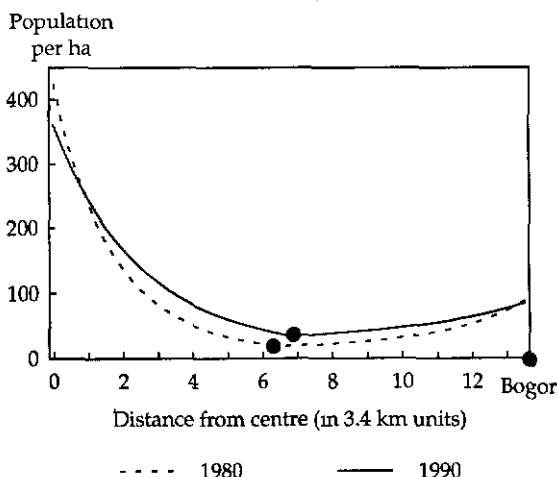
Despite the initial appearance of major intra-city thoroughfares and large parks and squares, Jabotabek has a high population density, especially within DKI Jakarta. In 1980, density near the centre averaged around 42,000 residents per square kilometre. In 1970, with

FIGURE 1 *Population Density Gradient for Urban Jabotabek, Excluding Kotamadya Bogor Segment^a*



^aExcludes segment of density gradient where density rises again near *kotamadya* Bogor.

FIGURE 2 *Population Density Gradient for Urban Jabotabek, Including Kotamadya Bogor Segment^a*



^aThe heavy dots in the figure indicate where the gradient switches from negative to positive slope, and where Bogor is located.

populations similar to that of Jabotabek in 1980, metro areas such as São Paulo, Buenos Aires and Mexico City had centre city densities of 18,000, 33,000 and 44,000 respectively (Ingram and Caroll 1981). Jabotabek is in the unenviable position of mimicking Mexico City.

While high for a large metropolitan area in a developing country, Jabotabek's population density at the centre is declining. Figures 1 and 2 plot standard 1990 and 1980 population density gradients for Jabotabek, depicting exponential relationships from regression results reported in table 1. The gradients are estimated for the 78 *kecamatan* (based on 1990 definitions) of urban Jabotabek, as listed in appendix A. The results for *kotamadya* Bogor are excluded from figure 1, in order to show the shape of the gradient without the effect of Bogor as a second centre of population pull. In figure 2, for the interested reader, we add the *kotamadya* Bogor component back into the density gradient, relevant to the extension of gradients into Bogor. About halfway to *kotamadya* Bogor the gradient starts to rise again, although at a quite modest rate.

Population density in table 1 and figures 1 and 2 is the average number of residents per hectare in each *kecamatan*. The estimating equation for each year is:

$$\ln(\text{pop. density}) = C_0 + b_0 \text{ distance} + b_1 \text{ distance squared} + b_2 \text{ Bogor dummy} + b_3 \text{ distance to Bogor (for Bogor kecamatan)} \quad (1)$$

The first three terms are standard in density analysis and apply to all *kecamatan*. The last two apply only to *kecamatan* in Bogor, and represent the fact that *kotamadya* Bogor (at the bottom of map 1) is a traditional city with its own pull, or centre of gravity. In table 1, we also report results for the density equation if the Bogor component and observations are dropped. The general patterns are similar, with and without Bogor. In equation (1), 'distance' is the straight-line distance from Monas (the traditional centre of Jakarta) measured in units of 3.4 km. The explanatory power of these simple regressions for 1980 and 1990 is very high. (The R^2 for 1980 is 0.71; for 1990 it is 0.73). The curves display the expected characteristics.

For 1980, density declines rapidly with distance from the centre of Jakarta. Since access to the city centre is at a premium, population generally clusters there, driving up land rents and density. As distance from the centre increases, density and land rents decline, since there is less demand for these locationally inferior sites. In practice, land price or rent gradients have the same general shape as density gradients, and those for Jakarta presented in Dowell and Leaf (1991) are no exception.

TABLE 1 *Population Density Gradient Regressions^a*
(standard errors in parentheses)

	ln (Residential population/hectare)			
	1980		1990	
	(1)	(2)	(3)	(4)
Constant	6.05* (.187)	6.27* (.210)	5.89* (.137)	5.94* (.145)
Distance	-.608* (.063)	-.760* (.094)	-.407* (.046)	-.448* (.065)
Distance squared	.019* (.004)	.036* (.009)	.009* (.003)	.015* (.061)
Dummy = 1 for kecamatan in Bogor	3.12* (.479)		2.36* (.352)	
Slope dummy distance from Bogor (only for kecamatan in Bogor)	-.448* (.075)		-.357* (.055)	
N	78	60	78	60
Adj R ²	.71	.73	.73	.75

*Significant at 5% level.

^aColumns (1) and (3) show estimates for equation (1) for the full urban area, including the second centre of gravity, Bogor. To show that inclusion of Bogor is not driving results, columns (2) and (4) drop the 18 *kecamatan* in the Bogor portion of Botabek.

Between 1980 and 1990 in Jabotabek there were major transport improvements with the construction and extension of intra-city highways, and there was massive population growth. In the absence of population growth, improved transport facilities reduce the demand to live at the city centre and the premium on location near it. Economic theory tells us that population density gradients then rotate anti-clockwise, becoming less steep—declining near the centre and rising near the fringes. On the other hand, population growth, in the absence of

transport improvements, shifts up the demand for space and the density gradient everywhere.

The 1990 and 1980 population density gradients in figure 1 reflect both transport improvements and population growth. The former work to rotate the gradient so it is less steep and density at the city centre is lower. At 3.4 kilometres from Monas, population density declined at a rate of 12% per kilometre in 1990, down from a rate of decline of 17% per kilometre in 1980. (These figures account for both the quadratic shape of figure 1 and the use of 3.4 km as the unit of distance in the regression equation.) In the absence of population growth, with rotation of the gradient due to transport investments, density at the city centre would have declined from 42,000 per square kilometre in 1980 to about 20,000 in 1990.³ But the 75% increase in the population of urbanised Jakarta between 1980 and 1990 (World Bank 1993) caused the gradient to shift up to its position in figure 1. While density did decline at the centre, over most of the metro area there was strong growth in population density (densification) further from the centre even within DKI Jakarta. Note that in figure 2 the 1980 and 1990 Bogor portions of the density gradient do not cross, reflecting Bogor's stagnant development.

Jabotabek's density gradient remains somewhat steep. The rate of population density decline per kilometre is in excess of, say, 11%,⁴ and the semi-log regression has a strong quadratic (as opposed to standard linear) shape. This feature reflects Jabotabek's continuing under-investment in urban transportation, particularly the absence of any mass transit system, poor management of vehicular traffic, and uncoordinated and poorly managed facilities that strongly discourage non-vehicular traffic. Further investments would flatten Jabotabek's density gradient, lowering population density near the centre to or below, say, 25,000 residents per square kilometre, and raising density on the fringes.

Future transport investments, along with rising per capita incomes, will also precipitate dramatic changes in land use. Three aspects stand out. Underlying the declines in density near the centre will be

³This is a 'back of the envelope' calculation, based on integrating population density over the metropolitan area to obtain total population, using the 1990 slope and calculating the constant term in the density equation consistent with the total 1980 population.

⁴This or a flatter slope is typical of São Paulo, Buenos Aires or New York City of 25 years ago. All evidence indicates further flattening of density gradients for these metro areas since then

conversion of many types of residential land to commercial use and, at the high end, residential market construction of higher rise apartments in particular neighbourhoods (e.g. Jakarta Selatan and perhaps one day the harbour). General declines in neighbourhood population densities will reflect rising incomes and gentrification in kampungs and traditional neighbourhoods, as upscaling consolidates land uses and increases floor space of residences. Finally there will be densification of land use further from the centre, as additional dwelling space is added to housing units and compounds.

Density changes have direct and dramatic consequences for aspects of urban development. High density accentuates negative spatial externalities, as the noise and waste products of land users more directly impinge on their neighbours. For example, Jabotabek residents are almost universally served by water pumped from the ground, rather than piped water connections, and by septic systems rather than sewer lines. High population densities (for single family homes, under 500 square metres of land) imply septic systems will generally be in 'failure' (Newby 1991), so that waste water will both rise to the surface, imposing general health risks, and combine with ground water to pollute pumped water. This is a well known problem with serious health implications for Jakarta residents (World Bank 1993). What is less well understood is how rapid population density changes can alter the situation. As central city densities decline some of the contamination problems will be alleviated. But more critically, as Jabotabek moves to invest more in sewers or off-site servicing of domestic waste, the natural sequence of neighbourhood redevelopment affects the optimal timing and placement of such investments. There is no point in providing trunk sewerage in old dense neighbourhoods today, if in a few years the neighbourhoods will be redeveloped and desired sewer configurations altered. Major investments in local roads, sewers and water mains should be timed to follow the natural neighbourhood redevelopment process.

Other Development Issues

Apart from these basic density-land use change questions, aspects of developing land use in Jabotabek deserve comment.

Suburban Sprawl. First, in juxtaposition to the high population density issues in DKI Jakarta, in the fringe areas of Botabek there are serious problems of 'urban sprawl' (Struyk *et al.* 1990; Aksoro 1994). Development occurs in 'pockets' of uncoordinated projects along major highways. Each pocket, the project of one private developer, is not linked to the next, so there are no networks of feeder and secondary

roads among developments. Pockets are not very deep, as developers rely on immediate access to the highway, rather than deepening their own developments through investment in feeder roads. The pattern of green spaces between pockets and between the ribbons of development along parallel highways is more a temporary accident than the result of long-term planning. Developments are poorly serviced with water connections, and sewage disposal still relies heavily on unregulated septic systems.

There is no pro-active local land use planning in Indonesia, and local rural officials in Botabek are not prepared for the impacts of the rapid development that follows tollway or arterial road construction. Preparation would involve: (1) forecasts of the local increase in demand for land for residential, commercial and industrial use, in advance of or in conjunction with tollway construction; (2) revision of local land use plans to meet and allocate this forecast demand and to identify and plan for consequent other infrastructure needs; and (3) constraint of developer proposals to fit with these revised land use plans and identified infrastructure needs. In practice, major land developers approach unprepared *kabupaten* officials with development proposals, location permits (i.e. development rights) are issued, and unprepared small current land owners are approached by the monopsonistic developer of their area to sell, so land may be assembled for development. Subsequently, so called national land use plans (held in a division of Public Works) are passively revised to reflect construction by these private developers, as sanctioned by the issue of location permits by local officials.

Land Tenure and Land Use Development. There is a fairly clear description of the complex system of land rights in Indonesia in chapter 5 of Struyk *et al.* (1990), and corroborating descriptions in Aksoro (1994) and Dowell and Leaf (1991). In urban, particularly kampung, areas, there are significant tracts of land for which those in possession do not have a registered claim of ownership, nor sound customary title evidenced by tax receipts and a letter from a local official. Squatters may try to establish stronger claims by starting to pay taxes on their land. Weaker claims, either without any sound customary title or with less than strong claims to customary title, may be sufficient for informal sales—transfers between two private parties that do not alter land use. However these weaker claims represent an extremely inhibiting force in the dynamics of urban development.

As Jabotabek continues to grow, to change industrial composition, to suburbanise industry, to decentralise and suburbanise population and to experience growth in real incomes and housing demand, there will be

continuing dynamic pressures to alter and upgrade existing land uses throughout much of the metro area. In large kampung neighbourhoods where weak land right claims are widespread, altering land use (to, say, commercial use or upscale housing) or upgrading land use (by consolidation of holdings into larger plots and dwellings) is not easy. It is difficult, for example, to assemble tracts of residential kampung land for conversion to commercial or lower density residential use. Those in possessory rights cannot sell their land (since they do not have title) and can only exercise their rights and realise some value to their land by continuing to reside on it, thus inhibiting changes in land use patterns. According to Dowell and Leaf (1991), strong claims raise land prices by 45% over weak claims. Those with possessory rights and weak claims will receive only a discounted price for their land if they 'sell' (with the land retaining the same residential use), and much less if the land is taken for the public interest (e.g. for housing projects financed by the government agency BTN) at compensation set by the government. The insecurity of the claim also itself affects quality of land use: those with more secure claims have a stronger incentive to invest in improving their homes and neighbourhood, without fear of losing their investments through poorly compensated takings or forced sales.

The straightforward way to solve the problem would be to devise a mechanism to grant registered claim status to squatters of some long standing.⁵ That would accelerate desired land conversion and upgrading, because these residents would now receive full prices if they sold their land for conversion and, otherwise, would have increased incentives to improve their properties. Such a mechanism would promote efficiency in the land market. But it has equity implications as well. It would increase the wealth of kampung residents and assure them of a greater share of the profits of redevelopment.

Socio-economic Stratification. At the neighbourhood level, land use in Jabotabek is mixed: residential and commercial uses exist side by side and small enclaves of high income residences are mixed in with low or middle income housing. However, Jabotabek as a whole does exhibit an expected general tendency to industrial clustering and to neighbourhood segmentation by socio-economic status. Generally we expect residents of the same socio-economic status to tend to group together, reflecting both whom they want as neighbours and their common need for income-appropriate neighbourhood shopping and service facilities.

⁵To discourage future (additional) squatting, new squatters should be routinely evicted.

TABLE 2 *Socio-economic Composition of the Kotamadya/Kabupaten of Jabotabek^a*

	Share of College Educated/Share of Population		Share of Those with Less than Primary School/Share of Population	
	1980	1990	1980	1990
Jakarta Selatan	2.0	1.8	.80	.58
Jakarta Timur	1.3	1.5	.63	.54
Jakarta Pusat	1.4	1.4	.69	.57
Jakarta Barat	1.0	1.1	.83	.73
Jakarta Utara	1.1	.87	.89	.75
Bogor	.24	.43	1.5	1.4
Bekasi	.33	.66	1.7	1.3
Tangerang	.40	.63	1.4	1.3

^aFigures are for population aged 10 years or older.

In Jabotabek in 1980 and 1990, Jakarta Selatan, Timur and Pusat were home to college educated residents (table 2): their share in the population of these regions relative to their share of the total population ranges from 1.3 to 2.0, depending on the area and year. In contrast, Jakarta Utara and the 'suburbs' in Botabek are home to less educated residents.

The trends in spatial socio-economic status movement from 1980 to 1990 are also interesting. Jakarta Selatan is losing some of its relative hold on the college educated, as these households suburbanise into high income residential pockets over the Bogor border, a trend that is likely to continue. But the low socio-economic status residents are also suburbanising. Table 2 indicates that shares of the least educated people relative to shares of total population fell everywhere. First, numerically this is possible.⁶ Second, it occurs here with relative shares in DKI Jakarta falling the most, as less educated people suburbanised.

⁶Consider three areas with shares in year 1 of less educated people and of population of respectively (.5, .4), (.2, .1), and (.3, .5). In year 2 these shares

In non-American cities around the world, suburbanisation of portions of both rich and poor into different residential communities is common. The flattening of density gradients applies to all income groups. Inner cities retain enclaves of poorer traditional and ethnic neighbourhoods, enclaves of traditional high income residents and newer developments for younger professional people. Suburbanisation of families, particularly in the child rearing ages, reflects a search for cheaper land and housing and also indicates suburbanisation of jobs and employment. All this appears to be happening in Jabotabek.

Suburbanisation involves extensive cross and circumferential commuting. People from the same household commute from suburban residential enclaves to different jobs in different locations: some in the inner city, some nearby, some in other suburbs. In Jabotabek significant commuting across *kabupaten/kotamadya* boundaries already occurs. Even in DKI Jakarta in 1990, 27–43% of a *kotamadya* resident workforce commuted outside their own *kotamadya*. From nearby areas of Botabek the percentages are much higher, with 50–62% of the local labour force commuting over their own *kabupaten* boundary into DKI Jakarta (GOI 1993, annexes). With further transport improvements and industrial and residential suburbanisation, this level of cross-commuting should rise throughout Jabotabek.

INDUSTRIAL COMPOSITION AND SPATIAL DISPERSION

Between 1980 and 1990, the composition of economic activity and employment in Jabotabek altered dramatically. Significant changes are given in table 3. The shares of manufacturing and financial and business services in total Jabotabek employment both increased by about 70%, the latter particularly in DKI Jakarta (see also table 4). This expansion occurred at the expense of community, social and personal (including government) services. The expansion in financial services is to be expected, given Jakarta's role as a financial and headquarters centre.

Spatial Distribution of Manufacturing

The spatial distribution of employment in Jabotabek has also changed significantly, with many implications for the dynamic development of land use in Jabotabek. The most marked feature is the rapid

change to (.6, .51), (.3, .16) and (.1, .33). The year 1 ratios of 1.25, 2, and .6 then all fall to 1.18, 1.88, and .3 respectively.

TABLE 3 *Industry Shares of Total Employment^a*
(%)

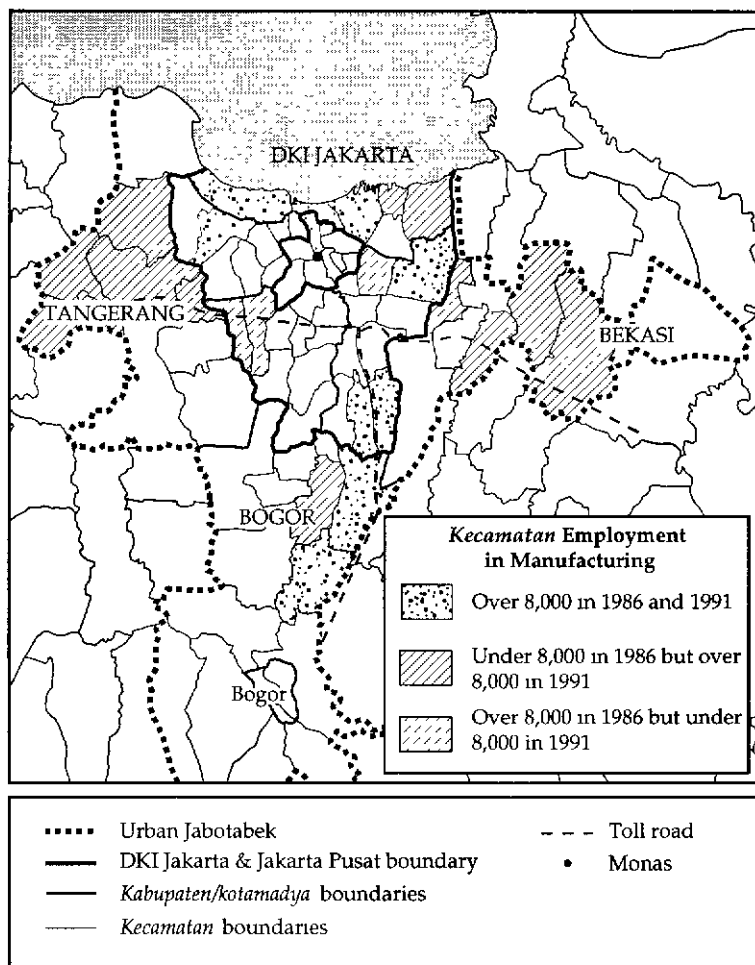
	Manufacturing	Financial and Business Services	Community, Social and Personal Services
Jabotabek			
1980	13	3	34
1990	22	5	25
DKI Jakarta 1990	20	7	29

^aFor all workers 10 years or older who worked the previous week.TABLE 4 *Suburbanisation of Manufacturing^a*

	1986	1991	1989-1991
% of formal sector Jabotabek employment in Botabek	43	56	na
% of Jabotabek employment in small firms in Botabek	24	na	na
% of Jabotabek plant births in Botabek for 1989-91			
Corporate sector	na	na	73
Unincorporated (but formal) sector	na	na	32

^aFigures relate to *kecamatan* for the 1990 urbanised portions of Jabotabek, using 1986 *kecamatan* boundaries.

Source: Revised DSP data files for the Annual Survey of Medium and Large Scale Manufacturers. The number for small firms in Botabek is from the 1986 Economic Census.

MAP 2 *Employment in Jabotabek*

Source: BPS Annual Survey of Medium and Large Scale Manufacturing as reprocessed and updated by DSP.

suburbanisation of manufacturing with the extensions of the toll roads into Bekasi and Tangerang in the late 1980s. The response within the five-year period containing the toll road development is given in table 4. Between 1986 and 1991 in the formal sector (medium and large size establishments with over 20 employees), the share of Jabotabek's manufacturing employment located in Botabek rose from 43 to 56%. Within the formal sector, for 1989–1991, 73% of all plant births in Jabotabek in the larger corporate sector (as opposed to unincorporated sector) occurred in Botabek. This happened despite the EPZ (export processing zone) status and protection enjoyed by the DKI Jakarta *kecamatan* of Cakung and Cilincing.⁷

The changing spatial distribution of manufacturing is pictured in map 2. The stippled areas represent *kecamatan* where formal sector manufacturing employment exceeded 8,000 in both 1986 and 1991.⁸ All these areas grew in employment between 1986 and 1991, by 40–120% depending on the *kecamatan*. The heavy slashed areas represent *kecamatan* whose formal manufacturing employment was less than 8,000 in 1986, but greater than 8,000 in 1991. Growth in all these *kecamatan* was extraordinary: the slowest *kecamatan* in the group had a 210% increase in manufacturing employment. But for the most eastern and western *kecamatan*, one grew fourteen-fold to 93,000 and three others grew from zero employment in 1986 to 13,000, 16,000 and 22,000 in 1991. Finally, the areas marked by broken slashes represent DKI *kecamatan* nearer the centre, where manufacturing employment exceeded 8,000 in 1986 but fell below 8,000 by 1991, generally with very large declines.

The suburbanisation of industry in rapid response to tollway development reflects the 'pent-up demand' of industrial establishments to escape the high costs of land and labour in DKI Jakarta, and to obtain large plots of land for development. Suburbanising in order to lower production costs helps manufacturers remain competitive in world markets. Since 1991, industrial development has spread rapidly further east and west into the *kabupaten* of Serang and Karawang beyond Tangerang and Bekasi, as is apparent from the August 1994 land use documents deposited in the Ministry of Public Works. This very rapid decentralisation along transport corridors suggests that, with further transport and communication development throughout Java, industry

⁷At the eastern end of Jakarta Timur and Jakarta Utara respectively.

⁸The cutoff figure of 8,000 is used because there are almost no areas with employment between 6,000 and 8,000.

could decentralise into interior medium and smaller cities beyond the Jabotabek area. Average wage costs for workers of similar skill are typically 100–150% higher in the largest than in the smallest metro areas, simply reflecting cost-of-living differentials (Henderson 1991). Even with decentralisation into Botabek, average labour costs in urban West Java remain much higher than in cities in other parts of Java, with the differential growing.⁹ Within Jabotabek itself there is a 25% wage differential between formal sector manufacturing in Jakarta and in Botabek. For standardised manufacturing processes, this remains a powerful incentive to decentralise, following enabling conditions such as appropriate infrastructure investments (transport and communications) and decentralisation of financing.

Multi-Centred Development

For Jabotabek this rapid decentralisation of industry has changed Jakarta from a metropolitan area dominated by one 'business district' for manufacturing (Jakarta Utara plus portions of Jakarta Pusat) to a distinctly multi-centred metro area. Statistically, for example, a formal sector manufacturing employment density gradient (following the figure 1 methodology, substituting employment counts for resident counts) for 1986 has a distinct monocentric shape, given in columns (1) and (3) of table 5. In table 5, density is the average number of manufacturing workers per hectare in a *kecamatan*. The density gradient has the same shape as in equation (1), except that we have dropped the quadratic term. This equation for 1986 has significant coefficients and an R^2 of 0.38. Graphically it has the same shape for 1986 as the population density gradients in figure 1. However, by 1991 employment density has changed so that the slope of the gradient is much flatter and central employment per hectare is halved. Moreover, the 1991 coefficients are insignificant and the R^2 for the 1991 equation is only 0.08. Thus there is no significant correlation between distance from the centre and employment density in 1991, which indicates the loss of monocentricity. The loss occurs regardless of specification. For example, dropping Bogor observations and variables yields almost identical results, as shown in columns (2) and (4).

These statements concerning the loss of monocentricity for manufacturing employment also apply to individual industries. We

⁹Source: Unpublished graphs of wage increases by occupation for the last four years, prepared by the Development Studies Project (DSP).

TABLE 5 *Employment Density Gradient^a*
(standard errors in parentheses)

	Ln (Manufacturing employment/hectare)			
	1986		1991	
	(1)	(2)	(3)	(4)
Constant	6.26* (.386)	6.45 (.366)	5.53* (.463)	5.55 (.510)
Distance (to Monas)	-.350* (.086)	-.404* (.087)	-.193 (.101)	-.198 (.116)
Dummy: <i>kotamadya</i> Bogor	3.07* (1.21)		1.55 (1.46)	
Slope dummy: distance to Bogor	-.665* (.176)		.393 (.230)	
Adj R ²	.382	.330	.078	.032
N	61	43	78	60

*Significant at 5% level.

^a*Kecamatan* definitions change from 1986 to 1990/91, with the splitting of some 1986 *kecamatan* into two or three. Columns (2) and (4) drop *kecamatan* in Bogor.

examine the three main industries—textiles and apparel (including footwear), chemicals (including petrochemicals), and machinery (including electrical machinery and transport equipment), which account for respectively 41%, 17% and 19% of Jabotabek manufacturing employment (the next most important sector, food processing, accounts for 5%). In table 6, we give the employment density gradients for these three industries for 1986 and 1991. As for overall manufacturing, each industry displays significant monocentricity in 1986 and none in 1991. Slope coefficients go to zero in 1991 for the displayed results and also for results estimated omitting Bogor coefficients and observations.

This loss of monocentricity does not imply simple spatial dispersion of industry *per se*. Rather it reflects development of new manufacturing

TABLE 6 *Individual Industry Employment Density Gradients*
(standard errors in parentheses)

	1986			1991		
	Textiles and Apparel	Chemicals	Machinery	Textiles and Apparel	Chemicals	Machinery
Constant	4.3* (.46)	4.2* (.45)	4.0* (.45)	4.4* (.48)	2.8* (1.5)	5.2* (.70)
Distance	-.25* (.10)	-.37* (.10)	-.25* (.10)	-.197 (.101)	-.04 (.10)	-.07 (.15)
Bogor dummy	1.9 (1.5)	3.2* (1.4)	1.4 (1.4)	1.2 (1.5)	-.05 (1.5)	-2.8 (2.2)
Slope dummy for Bogor: distance to Bogor	-.43* (.21)	-.56* (.20)	-.35 (.21)	-.27 (.24)	-.15 (.23)	.35 (.34)
Adj R ²	.16	.26	.20	.06	.00	.07
N	61	61	61	78	78	78

*Significant at 5% level.

centres outside of the north Jakarta area. In fact the degree of *spatial concentration* of industry in Jabotabek *increases* from 1986 and 1991. To see this we construct a measure of concentration—the Hirschman–Herfindahl index (HHI). The HHI for each industry is the sum of squares of each *kecamatan*'s share of total Jabotabek employment, calculated for 1986 and 1991 for the 61 *kecamatan* defined on a 1986 basis (so we can compare 1986 and 1991 indices). If an industry has employment uniformly distributed over the 61 *kecamatan* its HHI would be .016 (1/61), whereas if employment is concentrated all in one *kecamatan* the industry's HHI has the value 1. More generally, increases in HHI reflect increased spatial concentration of industry.

HHIs for all manufacturing and for the three major industries are given in table 7. In all cases the HHI increases, although most modestly

for machinery, the already most concentrated industry. Thus despite the development of several new manufacturing clusters in Jabotabek, spatial concentration increases between 1986 and 1991. Concentration levels are least for textiles and apparel.

Apart from the traditional Bogor *kotamadya*, Jabotabek shows four areas of industrial activity: Jakarta Utara, Bekasi and its border with DKI, Tangerang and its border with DKI, and Bogor *kabupaten* along the marked toll road and adjacent (to the west) unmarked highway (map 2). However this notion of four centres is a little misleading for two reasons. First, the Bogor strip has heavy apparel and textile

TABLE 7 *Geographic Concentration for Manufacturing (HHI)*

	All Manufacturing	Textiles and Apparel	Chemicals	Machinery
1986	.0534	.0467	.0789	.0925
1991	.0585	.0567	.0863	.0929

concentrations, while the other areas also have heavy representations of machinery, transport equipment, and chemicals. Second, the other three centres really form a band running east-west through Jakarta Utara and extending along the tollways and unmarked highways. Given positive externalities in production through information spillovers, and given the need to minimise transport costs among establishments engaged in different stages of processing of the same final product, this spatial band of contiguous activity among these centres is not unexpected.

Informal Sector Development

This picture of decentralising industry is incomplete in one critical aspect. It relates only to medium and large-scale formal sector industry.

In 1990, in DKI Jakarta for example, we estimate that about 40% of manufacturing employment was in the informal sector,¹⁰ i.e. in establishments with less than 20 employees (*small establishments* of 5–20 employees and 'cottage industries' of less than 5 employees). Small sector employment is much less suburbanised. In 1986 only 24% of small establishment employment was in Botabek, compared with 43% for the formal sector (table 4). Even in the formal sector, the larger, more mobile corporate establishments have a greater tendency to suburbanise than unincorporated establishments. In 1989–91, 73% of new plant births in the Jabotabek corporate sector were in Botabek, compared with 32% for the unincorporated (but formal) sector (table 4).

These numbers suggest that Jakarta Utara remains a key industrial centre, bolstered by the extension of the EPZ (export processing zone) from Cakung into Cilincing. Its importance extends beyond its 25% share of metro area formal sector manufacturing employment, because it includes heavy concentrations of small-scale and cottage industry. This smaller-scale industry can be important for two reasons. First, evidence from Rio de Janeiro and Calcutta (Mukherjee 1990; Sarkar and Mukherjee 1987) suggests that the small-scale sector has strong links to the corporate sector, supplying it with labour-intensive parts and components and filling orders for special feature or design components and machinery. The small-scale sector also plays a key role in assembling scrap and reprocessing it, in repairs, and in rebuilding of used parts and components. Second, according to evidence from Seoul (Lee 1989) and São Paulo (Hamer 1983), a different portion of the small-scale sector can play an incubator role. Small plants engage in speculative or experimental production, infusing manufacturing with new ideas and new adaptations of existing technology, and spurring competition with the corporate sector. If successful, these small plants grow into larger plants or are absorbed into the corporate sector. In both roles, the small-scale sector contributes significantly to overall manufacturing productivity growth, beyond simple measures of small-scale sector value added.

These considerations raise an important issue for the development of Jakarta Utara. Not only is large-scale manufacturing employment still

¹⁰This assessment is made by comparing 1990 employment totals for manufacturing from the Population Census and the Survey of Large and Medium Size Manufacturing Establishments. The 40% informal sector figure is comparable to figures from the 1986 Economic Census covering all industrial activity.

growing, but there remains a large concentration of small-scale industrial employment there. The DKI government has plans for a large mixed use waterfront development project to be built on landfill extending into the current harbour, as part of a general modernisation and gentrification of Jakarta Utara. Any planned waterfront development feeding into the overall development of Jakarta Utara should recognise the following:

- Jakarta Utara is a growing manufacturing centre.
- The small-scale sector in Jakarta Utara plays a critical role in manufacturing productivity and productivity growth, even if it appears not to be 'modern'. Attempts to eliminate small-scale activity will hurt formal sector productivity and competitiveness in world markets.
- Elimination of small-scale sector activity will create a spatial mismatch for kampung residents with limited resources for commuting between central area residential locations and suburban jobs for low skill workers. That could certainly foster kampung unemployment and dissatisfaction.

Location of Financial and Business Services

For non-manufacturing industries, spatial development is harder to track. The only useful source is the Economic Census held every 10 years (the last being 1986); its data are not readily available and remain largely unanalysed. The Population Census tells us the industry of residents according to where they live, not where they work, giving only a rough picture of industrial location.

Table 8 presents figures for employment in finance and business services based on the 1980 and 1990 Population Census. These figures indicate the continuing development of the 'Golden Triangle' of upscale business and financial services in Jakarta Pusat and running south to Block M in Jakarta Selatan. Along with these services is a high representation in Jakarta Selatan of social, community and personal services (35% of employment, compared with a Jabotabek share of 25%) and a high representation in Jakarta Pusat of retail, hotel and restaurant services (34% of employment versus 24% for Jabotabek). As in other metro areas around the world, upscale portions of financial, business, government and retail services are centralised.

TABLE 8 *Share of Employment of Residents in Finance and Business Services by Kotamadya/Kabupaten^a*
(%)

	1980	1990
Jakarta Selatan	4.8	8.4
Jakarta Timur	3.0	6.6
Jakarta Pusat	4.4	9.7
Jakarta Barat	3.5	6.5
Jakarta Utara	2.0	5.9
Bogor	1.0	1.2
Bekasi	1.2	2.8
Tangerang	2.0	3.0

^aFor all workers 10 years or older who worked in the previous week.

CONCLUSIONS

Jabotabek today has many attractive features. It is experiencing high levels of private investment in industrial and residential land uses; it has developed into a multi-centred metro area that is reducing its very high population densities near the city centre; it has green space spread throughout the metro area; it has the foundations for a strong intra-metro arterial highway system, and it radiates a vibrancy and a sense of a reasonable quality of life for many of its residents. As a large, rapidly growing metro area, however, it faces many challenges and changes.

- With further transport infrastructure investments in Jabotabek and further per capita income growth, there will be a continuing decline in population density nearer the core city, and densification and suburbanisation further out. The consequent rapid changes in land use patterns should influence the timing of other infrastructure investments (e.g. sewerage) in redeveloping areas of Jakarta.

- In this context of rapidly changing land use in Jabotabek, there is very little pro-active land use planning. Not only is there little planning to ensure reasonably efficient allocations of green space and configurations of industrial (obnoxious) and residential (non-obnoxious) land users, but developments are characterised by urban sprawl and poor infrastructure investments. Local officials permit ribbon (as opposed to deep-pocket) development, along highways, of projects with inadequate sewerage, local roads and water. Within Jakarta, efficient land redevelopment, especially in and around traditional kampungs, is hindered by a lack of clear title or land claim rights in lower income traditional areas. A simple solution would be to grant title to long-term residents with strong possessory rights.
- Industry is decentralising at an incredibly rapid rate in response to toll road construction. This reflects 'pent-up' demand to escape north-central Jakarta's high wage and land rent costs. Jabotabek has become a multi-centred metro area, rather than one dominated by central city employment, at least for manufacturing. This suburbanisation of industry takes the form of development of new centres or clusters of manufacturing activity, rather than simple dispersion. In fact spatial concentration of industry increased (modestly) between 1986 and 1991. The future promises more suburbanisation of industry and, potentially, decentralisation of manufacturing into smaller and medium sized cities in other parts of Java.
- Despite suburbanisation, Jakarta Utara remains a major corporate manufacturing centre, supported by a large small-scale manufacturing sector. Future waterfront developments should recognise the important role of small-scale manufacturing in that area of the city.

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APPENDIX A: URBANISED KECAMATAN OF JABOTABEK IN 1990/91¹¹

42 *kecamatan* of DKI Jakarta (excludes Kepulauan Seribu)

3 *kecamatan* of kotip Depok¹²

5 *kecamatan* of kotip Tangerang

4 *kecamatan* of kotip Bekasi

5 *kecamatan* of kotamadya Bogor

plus

for Bogor:

Cimanggis

Cibinong

Kedunghalang

Semplak

Bojong Gede

Sawangan

Ciampea

Ciomas

Cijeruk

Ciawi

for Tangerang:

Serpong

Ciputat

Pondok Aren

Curug

Teluknaga

for Bekasi:

Pondokgede

Cikarang

Cibitung

Tambun

¹¹Some of these have since been divided into two, and/or their boundaries redrawn.

¹²*Kotip* = *kota administratif* (lt. administrative city)—a recognised urban unit without the independence of a *kotamadya*